

MOSAIC: Data-Certified Stochastic Release under Structured Deployment Shift Supplementary Material

Anonymous submission

1 Notation and Exact Support Functions

For one task label, let $P = (p_1, \dots, p_G)$ collect the conditional fine-token laws and let $M \in \text{Stoch}(K, L)$ be a source-blind release channel. For an attacker assignment $a \in [G]^L$, define

$$v_{M,a,s}(c) = \sum_{z=1}^L M(c, z) \mathbf{1}\{a(z) = s\}. \quad (1)$$

Every $v_{M,a,s}$ lies in $[0, 1]^K$. The balanced Bayes accuracy has the two equivalent forms

$$\begin{aligned} A(P, M) &= \frac{1}{G} \sum_z \max_s (p_s M)(z) \\ &= \frac{1}{G} \max_{a \in [G]^L} \sum_s p_s^\top v_{M,a,s}. \end{aligned} \quad (2)$$

The second identity follows because a finite-alphabet Bayes rule can be chosen deterministically at each released token.

For empirical law $\hat{p}$, radius ϵ , and bounded score v , the confidence-set support function is

$$\Phi(\hat{p}, \epsilon; v) = \max_{p \geq 0, \mathbf{1}^\top p = 1, \|p - \hat{p}\|_1 \leq \epsilon} p^\top v. \quad (3)$$

Introducing positive and negative displacement variables gives a linear program. Its compact dual is

$$\min_{\nu, \lambda, \theta} \quad \nu + \hat{p}^\top \theta + \epsilon \lambda, \quad (4)$$

$$\text{s.t.} \quad \nu + \theta_c \geq v_c \quad \forall c, \quad (5)$$

$$-\lambda \leq \theta_c \leq \lambda \quad \forall c, \quad \lambda \geq 0. \quad (6)$$

Strong duality holds because the primal confidence set is nonempty and compact. The implementation also evaluates Eq. (3) independently by moving at most $\epsilon/2$ probability mass from low-score to high-score coordinates, respecting simplex boundaries.

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2 Adaptive Confidence Envelope

Let $\mathcal{C}_s = \{p \in \Delta_K : \|p - \hat{p}_s\|_1 \leq \epsilon_s\}$ and define

$$\bar{A}(\hat{P}, M) = \frac{1}{G} \max_{a \in [G]^L} \sum_s \Phi(\hat{p}_s, \epsilon_s; v_{M,a,s}). \quad (7)$$

Theorem 1 (Exact adaptive universal-attacker envelope). *On the event $\mathcal{E} = \{p_s \in \mathcal{C}_s \forall s, y\}$, $A(P, M) \leq \bar{A}(\hat{P}, M)$ simultaneously for every stochastic release channel M , including one selected from the same table. For fixed M , $\bar{A}$ is the exact supremum of $A(P, M)$ over $\mathcal{C}_1 \times \dots \times \mathcal{C}_G$.*

Proof. Using Eq. (2), take the supremum over the product confidence set. The assignment family is finite, so the supremum and assignment maximum commute. For a fixed assignment, the objective is additive in sources and the confidence set is a Cartesian product, so each source maximization separates and equals the support function in Eq. (3). This proves exactness for every fixed M . On $\mathcal{E}$, the true law is one feasible point in every confidence set, giving the upper bound. The statement is pointwise over the complete channel space, so it remains true after any measurable same-table selection of M . $\square$

Corollary 2 (Unconditional and label-aware attackers). *Let $X = (Y, C)$ and let r_s denote the natural, unbalanced law $P(X \mid S = s)$. Define the lifted channel $\tilde{M}((y, c), z) = M(c, z)$. Applying the theorem to $(r_1, \dots, r_G)$ certifies every source attacker observing Z without conditioning on Y . Replacing the output by (Y, Z) and using $\tilde{M}((y, c), (y', z)) = \mathbf{1}\{y = y'\} M(c, z)$ certifies the stronger attacker that also observes the true label.*

The corollary requires confidence regions for the natural joint laws and is not implied by source-label-balanced sampling. The primary experiments instead register the maximum label-conditional advantage, which removes the prevalence channel by design.

For conditionally i.i.d. categorical observations within each stratum, positive fixed count $n_{s,y}$, and an outcome-independent allocation $\delta_{s,y}$, the confirmation uses

$$\epsilon_{s,y} = \min \left\{ 2, \sqrt{\frac{2 \log((2^K - 2)/\delta_{s,y})}{n_{s,y}}} \right\}. \quad (8)$$

A union bound gives $\Pr(\mathcal{E}) \geq 1 - \sum_{s,y} \delta_{s,y}$. A zero-count stratum receives radius two and is covered deterministically. The locked bridge protocol used “exchangeable” informally for randomized held-out rows from the same stratum; arbitrary exchangeability alone does not imply this concentration bound, and is not claimed here.

3 Data-Certified Shift Bridge

For each source and task label, let $\mathcal{C}_{s,y}$ be the reference confidence set above and let

$$\mathcal{D}_{s,y} = \{q \in \Delta_K : \|q - \hat{q}_{s,y}\|_1 \leq \xi_{s,y}\} \quad (9)$$

be the confidence set from a labeled target bridge fold. Define

$$h_{s,y}(w) = \max_{p \in \mathcal{C}_{s,y}} p^\top w, \quad \underline{q}_{s,y}(z) = \min_{q \in \mathcal{D}_{s,y}} q(z). \quad (10)$$

The coordinate lower bound is $-\Phi(\hat{q}_{s,y}, \xi_{s,y}; -e_z)$ and is therefore exact. For fixed y , solve

$$\begin{aligned} \hat{t}_y &= \max_{t, W} t \\ \text{s.t. } & W_{cz} \geq 0, \quad \sum_z W_{cz} = t \quad \forall c, \\ & h_{s,y}(W_{:z}) \leq \underline{q}_{s,y}(z) \quad \forall s, z, \\ & 0 \leq t \leq 1. \end{aligned} \quad (11)$$

This is an LP. In particular, each support constraint is represented by variables $(\nu_{s,z}, \lambda_{s,z}, \theta_{s,z})$ satisfying

$$\begin{aligned} \nu_{s,z} + \theta_{s,z,c} &\geq W_{cz} & \forall c, \\ -\lambda_{s,z} &\leq \theta_{s,z,c} \leq \lambda_{s,z} & \forall c, \\ \nu_{s,z} + \hat{p}_{s,y}^\top \theta_{s,z} + \epsilon_{s,y} \lambda_{s,z} &\leq \underline{q}_{s,y}(z), \quad \lambda_{s,z} \geq 0. \end{aligned} \quad (12)$$

Strong duality for Eq. (3) makes this epigraph exact.

Theorem 3 (Adaptive finite-sample bridge membership). *Suppose the reference and bridge sets jointly cover all $p_{s,y}$ and $q_{s,y}$. Any feasible (t, W) in Eq. (11), including its same-table maximizer, satisfies, for $t > 0$,*

$$q_{s,y} = tp_{s,y}T + (1-t)r_{s,y}, \quad T = W/t, \quad (13)$$

for one source-blind stochastic T and source-specific probability vectors $r_{s,y}$.

Proof. Coverage and feasibility give, for every source and output coordinate,

$$q_{s,y}(z) \geq \underline{q}_{s,y}(z) \geq h_{s,y}(W_{:z}) \geq p_{s,y}^\top W_{:z}. \quad (14)$$

Thus $d_{s,y} = q_{s,y} - p_{s,y}W$ is coordinatewise nonnegative. Since every row of W sums to t , $\mathbf{1}^\top d_{s,y} = 1-t$. If $t < 1$, set $r_{s,y} = d_{s,y}/(1-t)$; if $t = 1$, then $d_{s,y} = 0$ and any residual law may be used. Also, $T = W/t$ is row-stochastic. Every implication above holds simultaneously for every feasible (t, W) on one event, so choosing the maximizer from the same confidence tables costs no additional multiplicity. $\square$

Proposition 4 (Robust optimality of the bridge LP). *Over the Cartesian product of all reference and bridge confidence sets, Eq. (11) returns the largest retained mass for which one fixed source-blind transform certifies Eq. (13) uniformly over every law in that product.*

Proof. For fixed (t, T) , the decomposition exists for a particular (p, q) exactly when $q_{s,y} - tp_{s,y}T$ is coordinatewise nonnegative: its coordinates then sum to $1-t$ and can be normalized into a residual law. Requiring this for all (p, q) in the product region is equivalent to

$$\max_{p \in \mathcal{C}_{s,y}} t(pT)(z) \leq \min_{q \in \mathcal{D}_{s,y}} q(z) \quad \forall s, z. \quad (15)$$

The two extrema separate because the region is a Cartesian product. Substituting $W = tT$ gives exactly the constraints of Eq. (11); Eq. (12) introduces no relaxation. Maximizing t proves the claim. $\square$

Corollary 5 (End-to-end target-population certificate). *Suppose $\hat{t}_y > 0$ for every required label. Insert the singleton transform $\{\hat{T}_y\}$ and contamination $\hat{\eta}_y = 1 - \hat{t}_y$ from the bridge LP into the transform-exact within-label source-distinguishability and utility optimizer. On the same joint event, every same-table-selected release channel and decoder obeys its certified contracts on the bridge population. This remains simultaneous across task labels, registered utility thresholds, and finitely many released alphabet sizes when their underlying reference and bridge tables share the registered allocation.*

Proof. The bridge theorem establishes membership in the structured class used by the transform-exact source-distinguishability and utility theorems below. Those theorems are pointwise over every release channel and decoder on the same reference event. Intersecting with the bridge confidence event and substituting the selected singleton transform proves the result. Thresholds and alphabet sizes index decisions made inside that already covered collection; separately learned tokenizers or erasers still require familywise allocation across their distinct tables. $\square$

If a required bridge stratum has count zero, its honest set $\mathcal{D}_{s,y}$ is the full simplex and every $\underline{q}_{s,y}(z) = 0$. For any $t > 0$, the probability vector $p_{s,y}T$ has a positive coordinate, contradicting the corresponding bridge constraint. Hence $\hat{t}_y = 0$. This is the operational missing-source boundary used in the bridge confirmation.

4 Channel Capacity and Common-Shift Transfer

Define

$$\text{Cap}_G(M) = \frac{1}{G} \max_{a \in [G]^L} \sum_s \max_c v_{M,a,s}(c). \quad (16)$$

Lemma 6 (Exact arbitrary-shift capacity). $\text{Cap}_G(M) = \sup_{r_1, \dots, r_G \in \Delta_K} A(R, M)$.

Proof. Expand $A(R, M)$ by Eq. (2). For fixed assignment a , each source law r_s can concentrate all mass on a token maximizing $v_{M,a,s}(c)$. The source suprema separate and the finite maxima commute, yielding Eq. (16). $\square$

For two sources, uniform-prior Bayes accuracy is $(1 + \text{TV}(r_1 M, r_2 M))/2$. The diameter of the convex hull of channel rows is attained by two rows, hence $2 \text{Cap}_2(M) - 1 = \max_{c, c'} \text{TV}(M(c, \cdot), M(c', \cdot))$, the Dobrushin coefficient.

For common pre-release channel T , define

$$\rho_T(M) = \max_c \text{TV}((TM)(c, \cdot), M(c, \cdot)). \quad (17)$$

Lemma 7 (Approximate invariance transfer). *For every source experiment P ,*

$$A(P, TM) \leq \min\{\text{Cap}_G(M), A(P, M) + \rho_T(M)\}. \quad (18)$$

Proof. For source s , convexity of total variation gives $\text{TV}(p_s TM, p_s M) \leq \sum_c p_s(c) \text{TV}((TM)(c, \cdot), M(c, \cdot)) \leq \rho_T(M)$. The success probability of any attacker changes by at most this distance. Averaging over sources and maximizing the attacker gives the additive bound. The capacity bound follows by treating $p_s T$ as an arbitrary fine-token law and applying the preceding lemma. $\square$

5 Structured External Shift

For label y , suppose

$$q_s = t_y p_s T_y + (1 - t_y) r_s, \quad t_y \geq 1 - \eta_y, \quad (19)$$

with common $T_y \in \mathcal{T}_y$ and common retained mass t_y across sources. For a polytope with finite extremes $\mathcal{V}_y = \text{ext}(\mathcal{T}_y)$, define

$$\begin{aligned} \bar{A}_y^X(M) = \frac{1}{G} \max_{T \in \mathcal{V}_y, a \in [G]^L} \sum_s \Big[& \\ (1 - \eta_y) \Phi(\hat{p}_{s,y}, \epsilon_{s,y}; T v_{M,a,s}) & \\ + \eta_y \max_c v_{M,a,s}(c) \Big]. & \end{aligned} \quad (20)$$

Theorem 8 (Transform-exact post-selection certificate). *On $\mathcal{E}$, simultaneously for every selected M and every external law in Eq. (19), $A(Q, M) \leq \bar{A}_y^X(M)$. For fixed M , the right-hand side is the exact supremum over the product confidence sets, transform polytope, allowed retained masses, and residual laws.*

Proof. Expand balanced Bayes accuracy using Eq. (2). For fixed T, t , and attacker assignment, the source confidence sets and residual simplexes separate. Their support values are $\Phi(\hat{p}_{s,y}, \epsilon_{s,y}; T v_{M,a,s})$ and $\max_c v_{M,a,s}(c)$. The latter is no smaller because every coordinate of Tv is a convex combination of coordinates of v . The maximizing retained mass is therefore $t = 1 - \eta_y$. The resulting confidence support is convex in T , so its maximum over a polytope is attained at an extreme. All remaining maxima are finite and commute. This gives exactness for fixed M ; pointwise coverage over all channels preserves the bound after same-table selection. $\square$

For a transform family that cannot be enumerated, let $B_y(M) = \min\{\text{Cap}_G(M), \bar{A}_y(\hat{P}, M) + \rho_{\mathcal{T}_y}(M)\}$.

Corollary 9 (Capacity-transfer fallback). *On $\mathcal{E}$,*

$$A(Q, M) \leq (1 - \eta_y) B_y(M) + \eta_y \text{Cap}_G(M). \quad (21)$$

Proof. The retained experiment is bounded by Eq. (18), the adaptive envelope bounds its unknown reference term, and residuals are bounded by capacity. Since $B_y(M) \leq \text{Cap}_G(M)$, the largest value over $t_y \geq 1 - \eta_y$ occurs at $t_y = 1 - \eta_y$. $\square$

The exact confidence envelope is no larger than this fallback because it is the supremum over the same set. At zero confidence radius, it becomes the known-population external risk

$$\begin{aligned} \frac{1}{G} \max_{T \in \text{ext}(\mathcal{T}_y), a \in [G]^L} \sum_s \Big[& \\ (1 - \eta_y) p_{s,y}^\top T v_{M,a,s} & \\ + \eta_y \max_c v_{M,a,s}(c) \Big]. & \end{aligned} \quad (22)$$

used by the separately implemented synthetic evaluator.

6 Utility Certificate

Fix decoder $g : [L] \rightarrow [J]$, let $\ell_y(z) = \mathbf{1}\{g(z) \neq y\}$, and write $u_y = M \ell_y$. Define

$$\bar{E}_{s,y}(M, g) = \Phi(\hat{p}_{s,y}, \epsilon_{s,y}; u_y), \quad (23)$$

$$\kappa_y(M, g) = \sup_{T \in \mathcal{T}_y} \max_c \{0, \max_c ((TM - M) \ell_y)(c)\}, \quad (24)$$

$$U_y(M, g) = \max_c u_y(c). \quad (25)$$

Define the transform-exact value

$$\begin{aligned} \bar{E}_{s,y}^X(M, g) = \max_{T \in \mathcal{V}_y} \Big[& \\ (1 - \eta_y) & \\ \Phi(\hat{p}_{s,y}, \epsilon_{s,y}; T u_y) + \eta_y \max_c u_y(c) \Big]. & \end{aligned} \quad (26)$$

Corollary 10 (Transform-exact adaptive utility certificate). *On $\mathcal{E}$, every selected (M, g) satisfies $\text{Err}_{s,y}(Q, M, g) \leq \bar{E}_{s,y}^X(M, g)$, and the right-hand side is exact over the stated confidence ball and shift class.*

Proof. Repeat the transform-exact proof for bounded loss ℓ_y , without averaging over sources or maximizing an attacker assignment. $\square$

The capacity fallback is

Corollary 11 (Adaptive shifted-utility fallback). *On $\mathcal{E}$, every selected (M, g) satisfies*

$$\text{Err}_{s,y}(Q, M, g) \leq (1 - \eta_y) D_{s,y}(M, g) + \eta_y U_y(M, g), \quad (27)$$

where $D_{s,y} = \min\{U_y, \bar{E}_{s,y} + \kappa_y\}$.

Proof. Apply the structured-shift proof to the bounded loss ℓ_y . The common component differs from its reference expectation by at most the largest positive directional defect κ_y , while the arbitrary residual concentrates on a row with maximal error U_y . $\square$

The transform-exact utility value is never larger than this fallback.

The exact population utility risk used in the independent evaluator is

$$\max_{T \in \text{ext}(\mathcal{T}_y)} \Big[(1 - \eta_y) p_{s,y}^\top T M \ell_y + \eta_y \max_c (M \ell_y)(c) \Big]. \quad (28)$$

7 No-Free-Lunch Boundary

Theorem 12 (Unrestricted differential shift). *For $G \geq 2$, $\text{Cap}_G(M) = 1/G$ if and only if all rows of M are identical.*

Proof. Identical rows make every released source law identical. Conversely, if two rows differ, choose two residual source laws concentrated on those rows and place every remaining source on either row. The released source laws are not all identical, so uniform-prior Bayes accuracy is strictly above chance. Exactness of Eq. (16) completes the argument. $\square$

For two fine tokens c_0, c_1 with different task labels, suppose a decoder has row-wise errors at most e_0, e_1 . The decoder decision event occurs with probability at least $1 - e_0$ under $M(c_0, \cdot)$ and at most e_1 under $M(c_1, \cdot)$. Total variation is therefore at least $1 - e_0 - e_1$, giving

$$2 \text{Cap}_2(M) - 1 \geq 1 - e_0 - e_1. \quad (29)$$

Thus a nontrivial task contract is incompatible with chance-level source capacity under unrestricted differential shift.

8 Repeated Queries and Side Information

Proposition 13 (Persistent-query equivalence). *Suppose the runtime samples one token Z per immutable item and every later query returns that stored token. For any finite or countably infinite number of queries, the complete transcript has exactly the same source-inference Bayes accuracy as one observation of Z .*

Proof. Every persistent transcript has the form $(Z, Z, \dots)$. The transcript is a deterministic function of Z , while Z is recovered from its first coordinate. They therefore generate the same sigma-field. The classes of measurable source decisions based on either observation coincide, so their optimal balanced accuracies are equal. $\square$

Fresh independent draws are a different mechanism. For a registered budget r , define the product channel

$$M^{\otimes r}(c, z_{1:r}) = \prod_{j=1}^r M(c, z_j). \quad (30)$$

Proposition 14 (Exact bounded-query reduction). *An attacker observing r conditionally independent fresh releases sees exactly one output of $M^{\otimes r}$. Substituting this product channel into the adaptive, bridge, structured-shift, and capacity results gives an exact certificate for that query budget, without a union bound over queries.*

Proof. Conditioned on fine token c , the transcript probability mass function is the right-hand side of Eq. (30). Every theorem depends on the release mechanism only through this conditional channel. Replacing the output alphabet $[L]$ by $[L]^r$ therefore reproduces the transcript experiment exactly. The alphabet has size L^r , which is a computational rather than a statistical restriction. $\square$

Proposition 15 (Unbounded fresh-query failure). *If two rows $M(c, \cdot)$ and $M(c', \cdot)$ differ, then their r -fold product laws have total variation tending to one. Consequently the binary source-inference capacity of $M^{\otimes r}$ tends to one as $r \rightarrow \infty$.*

Proof. For distinct finite-alphabet laws u and v , their Hellinger affinity $\alpha = \sum_z \sqrt{u(z)v(z)}$ is strictly below one. Product affinity equals α^r , and $\text{TV}(u^{\otimes r}, v^{\otimes r}) \geq 1 - \alpha^r \rightarrow 1$. Concentrating two source laws on fine tokens c and c' makes binary balanced Bayes accuracy $(1 + \text{TV})/2 \rightarrow 1$. The capacity is at least this value and at most one. $\square$

The registered attacker observes the persistent token, knows the complete mechanism, and has no additional item-level side information. If a variable H is also public, the relevant experiment is (Z, H) , not Z ; it must be included in the fine/output alphabet and certified jointly. In particular, original features, source-dependent metadata, mutable identifier behavior, timing, and private mechanism state are outside the theorem. A production runtime must durably bind each versioned item identifier to one immutable fine token and one sampled output; rebinding or silent rerandomization violates the registered mechanism.

Persistence is an item-level statement, not a population-level composition theorem. Tokens from distinct items associated with one source form a joint experiment. Even when each item is persistent, independent observations from different items can increase source distinguishability, while correlations, adaptive mechanisms, stopping times, and observable query counts can change the joint law further. Such access requires a certificate for the complete multi-item transcript. The product-channel proposition applies only to repeated conditionally independent draws from one fixed fine token under one fixed channel and a noninformative registered query count.

9 Missing-Source Non-Identifiability

Fix two required sources and a released-token law $q_0 M$ for source zero. Suppose an external audit contains no source-one rows, so its observed-data distribution is identical for every $q_1 \in \Delta_K$. Define

$$A_{\text{miss}}(q_0, M) = \sup_{q_1 \in \Delta_K} A((q_0, q_1), M). \quad (31)$$

Theorem 16 (Missing-source non-identifiability). *Fix q_0, M , and τ independently of the missing-source law and the audit randomness. Let a possibly randomized protocol observe no source-one rows and claim $A((q_0, q_1), M) \leq \tau$. If its false-certification probability is at most $\delta < 1$ uniformly over every q_1 , then for $\tau < A_{\text{miss}}(q_0, M)$ it can issue that claim with probability at most δ .*

Proof. The finite-simplex supremum is attained; call a maximizer q_1^* . The audit's observed-data distribution does not depend on the missing law. Consequently, every event on which the protocol claims a bound below $A((q_0, q_1^*), M)$ is a false-certification event under q_1^* , and uniform validity limits that event to probability δ . $\square$

The same indistinguishability argument for a missing source-label utility stratum replaces Eq. (31) by $\max_c (M \ell_y)(c)$. These are precisely the full-simplex terms used when a stratum has count zero. Moreover, if M is non-constant, then $A_{\text{miss}}(q_0, M) > 1/2$ for every q_0 : the mixture $q_0 M$ cannot equal every nonidentical channel row, and a

missing source concentrated on a differing row has positive total-variation distinguishability. Missing required strata must therefore be reported as unestimable unless an external structural assumption identifies their laws.

10 Abstention Theory

Let $V(P, e)$ be the globally optimal certified worst conditional error for confidence balls centered at P with common radius e . With deployed radius e_n , define $r_n^* = \sup\{r \geq 0 : V(P, e_n + r) \leq \tau_U\}$.

Proposition 17 (Finite-sample abstention envelope). *For H registered strata of size n ,*

$$\Pr(\text{MOSAIC abstains}) \leq \min\{1, H(2^K - 2)e^{-n(r_n^*)^2/2}\}. \quad (32)$$

Proof. If every empirical stratum is within distance r of its population law, each radius- e_n ball around the empirical law is contained in the radius- $(e_n + r)$ ball around the population law. A feasible population-centered pair for the larger balls remains feasible for the empirical problem. Apply the Weissman bound and a stratum union bound, then let r approach r_n^* . $\square$

For the preregistered local power curve, write $V_n(P) = V(P, e_n)$. Under local uniqueness of the decoder and certificate branches together with strong regularity of the resulting parametric linear program, the value is differentiable. If $h_{n,s,y}$ is its simplex gradient and $\Sigma_{s,y} = \text{diag}(p_{s,y}) - p_{s,y}p_{s,y}^\top$, the finite- n delta-method prediction is

$$\Phi_N \left(\frac{\tau_U - V_n(P)}{\sqrt{n^{-1} \sum_{s,y} h_{n,s,y}^\top \Sigma_{s,y} h_{n,s,y}}} \right). \quad (33)$$

The locked analysis estimates each gradient by centered simplex-preserving finite differences at step sizes 10^{-4} and 5×10^{-5} and rejects the approximation if their maximum discrepancy exceeds 5×10^{-3} .

11 Optimization Certificate

For completeness, fix a decoder g and write $b_{y,T,a,s}$ for the support-function epigraph with score $Tv_{M,a,s}$, $c_{a,s} \geq v_{M,a,s}(c)$ for every c , and $d_{s,y,T}$ for the utility support epigraph with score Tm_{ℓ_y} . Each b and d is represented by the dual in Eq. (6); its score is linear in M . The transform-exact LP is

$$\begin{aligned} \min_{M, \zeta} \quad & \zeta \\ \text{s.t.} \quad & M_{cz} \geq 0, \quad \sum_z M_{cz} = 1 \quad \forall c, \\ & \frac{1}{G} \sum_s [(1 - \eta_y)b_{y,T,a,s} + \eta_y c_{a,s}] \leq \tau_{P,y}^{\text{BA}} \quad \forall y, T, a, \\ & (1 - \eta_y)d_{s,y,T} + \eta_y u_y^{\max} \leq \zeta \quad \forall s, y, T, \end{aligned} \quad (34)$$

where $u_y^{\max} \geq (M\ell_y)(c)$ for every c and $\tau_{P,y}^{\text{BA}} = G^{-1} + (1 - G^{-1})\tau_{P,y}$. All epigraph variables are minimized indirectly or

upper-constrained, so an optimum can take each at its smallest valid value. Enforcing every finite transform extreme and all G^L attacker assignments exactly implements the maxima in the certificates. Enumerating all J^L decoders and taking the smallest objective returns the global transform-exact finite-alphabet solution.

For the capacity fallback, each expression $\min\{\text{Cap}_G(M), \bar{A}_y + \rho_y\}$ and $\min\{U_y, \bar{E}_{s,y} + \kappa_y\}$ is represented by a binary disjunction. With branch bit b , the first branch constraint is relaxed by $2b$ and the second by $2(1 - b)$; all constituent quantities lie in ranges for which two is a valid big- M constant. The implementation analogously disjoins the utility branches, then enumerates decoders. Thus the fallback is a finite MILP, not a local nonlinear search.

Globality is asserted only when every decoder and attacker assignment is enumerated, the exact LP reports an optimal status or the fallback MIP gap is at most 10^{-10} , and the solver feasibility tolerance is 10^{-9} . The original bridge run preserves raw solver outputs, but claim-grade decisions come from a strict replay triggered by an adversarial proof audit. It contracts retained mass using a 10^{-9} outward guard until every independently recomputed membership slack is non-negative, optimizes source constraints with a 10^{-6} inward margin, adds 10^{-9} to recomputed risk bounds, and compares those outward values with the original thresholds using zero decision tolerance. A timeout, negative membership slack, oversized assignment family, or post-solve mismatch can only abstain. The disclosed v2 correction to this repair is detailed below.

The exact audit checks zero-radius agreement with a separate population evaluator, binary confidence-ball endpoints, 100 randomized dominance comparisons with the fallback, and dense channel grids. The fallback audit compares 30 randomized small instances with exhaustive 21^K grids and includes a witness in which the best stochastic certified error is 0.371588 while every deterministic release has certified error 1.

12 Locked Experimental Protocol

The synthetic population has two labels, two sources, and three fine tokens. For label zero the source laws are $(0.80, 0.15, 0.05)$ and $(0.65, 0.30, 0.05)$; for label one they are $(0.05, 0.20, 0.75)$ and $(0.05, 0.35, 0.60)$. The common-transform set is the convex hull of identity and a nearest-neighbor smoothing channel. The hard safety scenario uses $\eta = 0.20$, source-advantage threshold 0.30, and worst conditional error threshold 0.40. The retention scenario uses $\eta = 0.10$, thresholds 0.35 and 0.45. Every source-label stratum contains exactly n independent draws.

The confirmation fixes 1,000 replicates in each of three hard-scenario and five retention-scenario sample-size cells. It compares eight rules and stores 64,000 method-level receipts. The preregistration fixes all seeds, methods, intervals, and pass criteria, hashes the theorem and implementation, and was pushed before the confirmation began. Pilot seeds zero through 199 are excluded. The oldest aggregate-only pilot is retained as design history but cannot support a channel-level claim; every later pilot row is replayed by the exact evaluator.

Matched finite-family baselines. A separately locked extension evaluates the same 1,000 paired tables per scenario with a 5^3 grid of binary-output stochastic channels and all 2^2 binary decoders, for 500 channel-decoder pairs. The confidence-region grid applies MOSAIC’s one simultaneous pre-channel event but restricts optimization to these 500 pairs. The deterministic baseline applies the same shift-aware certificate to every deterministic three-to-two map. We report it as a deterministic restricted-encoder baseline, not as FARE: official FARE trains a fair decision tree for demographic parity on a different raw-input protocol, whereas the present study evaluates within-label source inference under a pre-release geographic-shift contract. An end-to-end FARE number would not be commensurate with this contract.

The LTT baseline tests every fixed pair without reusing MOSAIC’s table region. Index its source-attacker and source-label utility inequalities by j . After removing the known contamination term, each has an empirical mean $\hat{\mu}_j$, unsafe null $H_j : \mu_j \geq b_j$, and effective sample size n_j (or Gn for a balanced attacker average over G source strata). Its one-sided Hoeffding value is

$$p_j = \exp\{-2n_j(b_j - \hat{\mu}_j)_+^2\}. \quad (35)$$

The unsafe candidate null is the union of its component nulls, so $p_{\text{cand}} = \max_j p_j$ is valid by an intersection-union test: whenever any H_j holds, rejection at level α implies rejection of that true component at level α . Holm step-down then controls familywise false certification across all 500 candidates at $\delta = .05$. Among rejected candidates, LTT selects the smallest point-estimated worst error with a fixed index tie break; no rejection means abstention. The population evaluator grades all four rules on the paired untouched law, including every abstention and false acceptance.

The data-certified bridge confirmation was locked separately after excluding all real-study seeds through 1199. It runs seeds 1200–1219 on Waterbirds, Camelyon17-WILDS, CivilComments-WILDS, BiasBios-Clinical, and GaitPDB. Within each represented target source-label stratum, a deterministic seeded split assigns two thirds to the bridge and one third to an untouched diagnostic set. The frontier has identity plus twelve settings from pinned official NLP, LEACE, R-LACE, TaCo, and MANCE++ repositories, with no proxy implementations. Each of 13 candidates has one reference and one bridge table; familywise $\delta = 0.05$ is divided across these 26 tables, and each table’s allocation is divided across its four source-label strata. Thus each multinomial radius uses $\delta/(13 \cdot 2 \cdot 4)$. The same event covers all bridge transforms, two- and four-token release optimizers, and utility thresholds $\{0.30, 0.35, 0.40, 0.45, 0.49\}$.

The primary rule uses $L = 2$, maximum source advantage 0.35, and maximum worst-stratum error 0.40. It selects the feasible candidate with minimum bridge-certified error and a lexical tie break. The minimum-error candidate is then re-optimized at $L = 4$ without inspecting diagnostics. All 100 dataset-seed jobs, 1,300 candidate rows, and 1,400 global optima are required; the stopping rule forbids outcome-based replacement or omission. Confirmatory gates require zero optimization errors, zero primary diagnostic contract violations, at least 16 estimable primary deployments, BiasBios

deployment on at least 16/20 seeds at threshold 0.40 and 14/20 at 0.35, pointwise noninferiority of the globally optimized $L = 4$ objective, and zero retained mass for every Camelyon job with missing bridge support. The claim is restricted to populations drawn i.i.d. from the same target stratum laws as the bridge fold and is not a clinical-safety claim.

After 27 of 100 raw receipts had been written, but before any receipt contents or outcomes were inspected, an adversarial proof audit identified that the original tolerance-based acceptance rule did not justify an exact software contract. A hash-locked numerical amendment therefore supersedes every raw bridge decision with the strict replay above. The amendment records the timing, the information seen (file-names, counts, timestamps, and process status only), all code hashes, the zero-tolerance decision rule, and the requirement to replay all 1,400 optima regardless of outcome. Both raw and strict receipts remain public so this correction cannot be hidden.

After the complete v1 strict audit, we found that its repair added the guard to the denominator of every output constraint, including a transform column that was identically zero. The exact constraint for such a column is $q_{s,y}(z) \geq t_y \cdot 0$ and cannot restrict t_y ; replacing the zero by 10^{-9} incorrectly collapsed retained mass to zero. All v1 outcomes had already been inspected and showed zero deployments. Before any corrected replay, a second hash lock disclosed that timing and froze a single change: omit the guard-ratio contraction only when every entry of the complete transform output column is exactly zero, then recheck every original slack, including that column. No table, candidate, threshold, radius, split, release optimizer, or diagnostic changed. The scope audit compares all 2,600 label certificates: all 918 changed labels have the required structural zero and no other label changes. Deterministic replay matches all 100 v2 receipts; exact rational arithmetic verifies nonnegative slack for all 1,300 bridges and outward bounds for all 1,400 releases. The v1 receipts remain part of the artifact and v2 is identified as post-audit corrective evidence.

13 Confirmation Receipts and Complete Primary Cells

The separate replay checks all 64,000 seeds and channels, 192,000 stochastic rows, 128,000 exact risks, 128,000 decisions, and 1,152 aggregate fields; the maximum discrepancy is zero in double precision. The release manifest contains the preregistration, report, and replay SHA-256 digests.

For zero events in 1,000 trials, the exact 95% Clopper-Pearson upper endpoint is 0.3682%. MOSAIC’s retention rate has interval $[53.16, 59.40]$; held-out certification has $[1.62, 3.67]$ and finite LTT has $[2.28, 4.60]$. The theory alignment audit links both pre-outcome and post-outcome receipts and passes with mean absolute deployment error 0.001968 and primary-cell error 0.004389. The finite-sample concentration abstention bound is vacuous through $n = 500$, then falls to 0.03097 at $n = 1000$ and 7.60×10^{-9} at $n = 2000$; we report that looseness rather than replacing it with the local approximation.

Rule	Hard safety boundary, $n = 125$			Retention regime, $n = 250$		
	Deploy	False	Safe	Deploy	False	Safe
Always deploy plug-in	100.0	57.6	42.4	100.0	0.0	100.0
Plug-in continuum	73.7	42.1	31.6	100.0	0.0	100.0
Shift-unaware MOSAIC	80.3	76.6	3.7	100.0	0.0	100.0
Held-out fixed	0.0	0.0	0.0	2.5	0.0	2.5
Finite LTT	0.0	0.0	0.0	3.3	0.0	3.3
Deterministic MOSAIC	0.0	0.0	0.0	0.0	0.0	0.0
MOSAIC	0.0	0.0	0.0	56.3	0.0	56.3
Population oracle	100.0	0.0	100.0	100.0	0.0	100.0

Table 1: All eight locked rules in both primary cells (rates in percent). The oracle verifies that a safe channel exists; it is not deployable.

Rule	Hard boundary		Retention	
	Deploy	False	Deploy	False
MOSAIC continuum	0.0	0.0	57.9	0.0
Holm LTT grid	0.0	0.0	28.3	0.0
Table-region grid	0.0	0.0	0.0	0.0
Deterministic adaptation	0.0	0.0	0.0	0.0

Table 2: Complete matched-baseline outcomes (percent of 1,000 paired tables per scenario). The LTT and table-region families contain the same 500 fixed channel–decoder pairs.

14 Matched Baselines and Bridge-Validity Stress

Matched baseline receipts. The schema-repaired report preserves every numerical field from its immutable predecessor. Its deterministic audit replays 2,000 tables, 8,000 method rows, 12,000 diagnostics, and all aggregate cells with no failure. This is a code-independent entry point and row-level recomputation, but not independent human review. All digests are in the release manifest.

MOSAIC’s exact 95% deployment interval is [54.77, 60.98] and Holm LTT’s is [25.53, 31.20]. Their discordances are 312 MOSAIC-only and 16 LTT-only, with 267 joint deployments and 405 joint abstentions. The paired difference is 29.6 points. The exact two-sided McNemar value, computed only after the locked audit, is 2.279×10^{-72} and is descriptive. Zero false acceptances in 1,000 trials has exact 95% upper endpoint 0.3682% for each method–scenario cell.

Bridge misspecification receipts. The audit replays 12,000 tables, 216,000 replicate fields, and 228 aggregate checks. It finds no failure of a claimed bridge implication on the simultaneous confidence event.

For the valid mechanism, the exact 95% intervals are [0, .368], [28.14, 33.97], [99.63, 100], and [99.63, 100] percent as n increases. For every invalid cell, the exact 95% upper endpoint is 0.3682%. This power transition is consistent with shrinking multinomial regions: evidence for a valid common transform becomes decisive, while the registered violations stay below the required retained mass.

15 Real Data-Certified Bridge Study

The 100-receipt manifest links the deterministic, exact-rational, correction-scope, and comparator audits. Their di-

gests are consolidated in the release manifest.

At the primary .40 threshold, the bridge plug-in violation interval is [6.20, 28.31] percent for 7/47, validation-only is [18.85, 43.21] for 18/60, and unconditional deployment is [36.21, 58.98] for 38/80. Strict MOSAIC’s 0/20 interval is [0, 16.84] percent, with one-sided 95% upper bound 13.91%. Its 20 selections are six INLP, five LEACE, and nine MANCE++; at .49, three Waterbirds jobs join the 20 Bias-Bios jobs. CivilComments and GaitPDB abstain throughout, and all Camelyon jobs certify zero retained mass because a required target source–label stratum is absent. This is the theorem’s missing-support case, whereas treating those 20 releases as safe would be statistically unidentifiable.

Dataset grouping sensitivity. The 100 rows are 20 randomized jobs on each of five shared benchmarks, not 100 independent deployment domains. At the primary threshold, every one of the 20 strict deployments is on BiasBios-Clinical (median retained mass .797; range [.756, .858]). Camelyon17 and GaitPDB have zero retained mass from missing support; CivilComments and Waterbirds have no jointly feasible strict source–utility release. A descriptive bootstrap that resamples the five datasets gives a strict deployment-rate interval [0, .60]. Leaving out BiasBios leaves zero strict deployments, whereas leaving out any other dataset leaves 20. These summaries describe concentration of the experiment and do not alter the per-job finite-sample guarantee.

Fresh corrected confirmation. Before any outcome from seeds 1220–1224, a separate lock fixed the same five datasets, 13 official candidates, contracts, and new bridge–diagnostic splits. Its strict-v2 manifest links the independent and exact-rational audits; their digests appear in the consolidated checksum index. All 25 raw jobs have 13 rows and zero optimization errors; strict replay reproduces 325 candidates and 350 global optimizations with no failure, and rational replay verifies every serialized bridge and release. At .40, it deploys on 5/5 BiasBios jobs with 0/5 observed diagnostic violations (three INLP and two MANCE++), and abstains on every one of the other 20 jobs. This is an untouched confirmation of the corrected pipeline, while again showing that current real releases concentrate on one benchmark.

The original v1 strict repair produced zero deployments at all five thresholds. The locked scope audit finds 918 zero-to-positive retained-mass corrections, all and only on label transforms with an identically zero output column. The

Target mechanism	Population min. t	$n = 500$	$n = 1000$	$n = 2000$	$n = 5000$	Invalid false
Common source-blind transform	1.000	0.0	31.0	100.0	100.0	–
Underdeclared contamination	.775	0.0	0.0	0.0	0.0	0/4,000
Source-specific transforms	.500	0.0	0.0	0.0	0.0	0/4,000

Table 3: Bridge acceptance rates (percent of 1,000 tables per cell) for the registered minimum retained mass $t \geq .80$. The two invalid mechanisms are specific stress tests, not an exhaustive misspecification class.

Rule	.30	.35	.40	.45	.49
Strict MOSAIC	20/0	20/0	20/0	20/0	23/0
Capacity transfer	0/–	0/–	0/–	0/–	19/0
Bridge plug-in	38/6	47/7	47/7	62/18	77/16
Validation plug-in	68/15	76/15	80/18	92/26	100/18
Always deploy validation	100/47	100/39	100/38	100/34	100/18

Table 4: Deployments/observed diagnostic violations across 100 real jobs at each worst-stratum error contract. Violation denominators include only estimable deployments. Validation-only and unconditional rules deploy 20 unestimable Camelyon jobs at every threshold; strict MOSAIC abstains on them.

largest restoration is .857614. Every corrected original inequality has nonnegative exact-rational slack. Thus v2 fixes a documented software over-abstention bug; it does not weaken the certificate or erase the v1 record.

16 Transform-Exact Confirmation and Audit Amendment

The strict-decision v2 preregistration, 10,000-row report, and independent audit are linked by the consolidated release manifest.

Declared numerical amendment. The original v1 confirmation passed every pre-audit gate, but its independent replay marked two of 10,000 deployment decisions as mismatches. All 80,000 recomputed certificate and external-risk quantities matched. The two returned capacity-fallback channels exceeded the source-leakage threshold by 4.12×10^{-10} and 4.40×10^{-10} : below the solver’s 10^{-9} feasibility and 2×10^{-7} post-hoc tolerances, but above the replay’s 10^{-10} decision tolerance. We preserved the failed audit, changed the deployment layer to recheck every source-leakage certificate at 10^{-10} , and recorded before rerunning that exactly those two rows should flip to abstention without changing any primary gate. The hash-locked v2 rerun matched that prediction exactly; its audit has zero mismatch.

The confidence event holds on 999 of 1,000 tables in the hard cell, on all retention $n = 125$ tables, on 999 tables at $n = 250$ and 500, and on all tables at $n = 1000$. There are no failures on the confidence event and no observed false acceptances off it. For zero events in 1,000 trials, the exact 95% binomial upper endpoint is 0.3682% per method-cell; this empirical endpoint is distinct from the theorem’s familywise $\delta = 0.05$ guarantee.

The exploratory five-threshold receipt reuses the audited channels and exact population risks, holds the source-leakage threshold at .35, and changes only the utility decision threshold. Because four thresholds were selected after outcomes, the sensitivity table in the main paper is explicitly descriptive. All 20 method-size-threshold cells have zero false acceptance, and exact retention is never lower than capacity

transfer.

17 Transform-Exact Verification Matrix

The exact support implementation is checked in three independent ways. At zero radius, source leakage, and utility match a population evaluator that enumerates every transform extreme, attacker assignment, and residual simplex vertex. For binary fine alphabets, confidence-ball endpoint enumeration reproduces the support-function result. Finally, 100 randomized instances verify that the exact envelope never exceeds the capacity fallback. The optimizer is then checked against dense channel grids and an independent post-solve certificate recomputation. Eleven focused transform-exact tests pass as part of a repository-wide run of 255 tests plus 14 subtests.

18 Current Real-Data Evidence

Registered 100-job study. The primary real-data study contains 20 locked seeds on each of BiasBios-Clinical, Camelyon17-WILDS, CivilComments-WILDS, GaitPDB, and Waterbirds. Each job evaluates the same 13-candidate family built with the official INLP, LEACE, R-LACE, TaCo, and MANCE++ implementations. The strict-v2 replay reproduces all 100 decisions; an exact-rational audit verifies 1,300 bridge certificates and 1,400 release bounds. Table 6 gives the aggregate comparison at the primary .40 worst-stratum error contract.

Strict MOSAIC releases 20 BiasBios-Clinical interfaces at every registered threshold from .30 through .45, and 23 interfaces at .49; every released job has an estimable untouched diagnostic and no observed diagnostic violation. The direct target-table comparator releases 36 jobs with no observed diagnostic violation. Its guarantee applies to the target table itself, whereas MOSAIC’s bridge program certifies a structured class of target laws. The bridge plug-in, which keeps the learned transform but omits the confidence event, releases 47 jobs and records seven diagnostic violations. This side-by-side comparison separates target-table certification from transport certification and quantifies the value of the

Scenario	n	Safe deploy (%)		Mean cert. error		Mean gain (pp)	False accepts
		transfer	exact	transfer	exact		
Hard boundary	125	0.0	0.0	.5004	.4959	0.45	0/2000
Retention	125	0.3	53.0	.4808	.4482	3.26	0/2000
Retention	250	57.8	99.9	.4479	.4070	4.09	0/2000
Retention	500	100	100	.4162	.3799	3.63	0/2000
Retention	1000	99.8	100	.3888	.3612	2.77	0/2000

Table 5: Every locked transform-exact confirmation cell. Each method has 1,000 paired tables per row. The exact objective is no larger on all 5,000 pairs and strictly smaller on all 4,000 retention pairs.

Rule	Released	Estimable	Diagnostic violations
Strict MOSAIC	20	20	0
Direct target-table certificate	36	36	0
Bridge plug-in	47	47	7
Validation-only selection	80	60	18
Unconditional selection	100	80	38

Table 6: Aggregate primary-contract outcomes over 100 registered jobs. The direct target-table rule certifies the sampled target table; strict MOSAIC certifies the registered bridge-admitted population class.

simultaneous regions.

California-to-Texas confirmation. Five fresh ACSIncome jobs use California as the reference population and Texas as the labeled target population. The task is income above \$50,000; binary sex is the audit source and is excluded from released features. Five locked seeds evaluate the complete official 13-candidate frontier. The independent strict replay checks all 65 candidate rows and 70 global optimizations. At the .40 primary contract, MOSAIC abstains on all five jobs. At the registered .45 and .49 contracts, it releases all five jobs: three R-LACE and two identity interfaces. The two identity rows are degenerate no-op releases and are not counted as evidence that editing helped. Every released interface satisfies its untouched Texas diagnostic. The best .40 errors range from .4023 to .4145. In the active worst stratum, the bridge-residual charge is .161–.211, while the finite-sample charge is at most 1.6×10^{-7} ; removing the residual charge would put every fold below .25. A post-review $K = 8$ extension on one previously studied seed retains only .377–.540 of the target laws. All 13 official candidates then reduce to a .50-error constant release and abstain.

High-support $K = 8$ extension. A separate post-review protocol fixes three employment transfers before execution and raises both certification tables from 6,000 to 16,000 balanced rows per source-label stratum (64,000 reference and 64,000 bridge rows per job). It evaluates the complete 13-candidate official frontier in every job, for 39 rows total. At the same .40 primary utility threshold, strict MOSAIC certifies a nonconstant release on Florida and New York; Illinois remains an abstention. The two released interfaces have safe untouched diagnostics and no contract violation in either 100-draw operational replay.

This is a measured support effect, not a broad claim that $K = 8$ is solved: the protocol covers three employment transfers at one locked seed, and one of them still abstains. The frozen audit checks all 39 candidate rows, every official-

Target	Strict selection	Diagnostic error	Operational violations
Florida	INLP, rank 8	.2978	0/100
Illinois	abstain	–	–
New York	MANC++	.2772	0/100

Table 7: Locked high-support $K = 8$ employment extension at $\tau_U = .40$. The certified error upper bounds are .3849 (Florida) and .3716 (New York), both below the contract.

Rule	Deploy	Violations	Abstain
Direct target table	36	16	0
Strict MOSAIC	20	0	16

Table 8: Outcomes on the 36 direct-selected jobs after moving to an exact member of each learned bridge class.

method provenance record, both sampling caps, and the 200 operational draws.

Admitted-shift stress. The direct target-table rule and MOSAIC first see the same stored real token tables. For each direct deployment, we then choose residual simplex vertices that maximize that release’s Bayes source attacker and form $q^* = t\hat{p}T + (1-t)r^*$. Reconstruction error is zero, so every q^* lies inside the candidate’s learned bridge class. All 36 lie outside the direct target confidence region. Table 8 reports the primary .40 outcome. This post-hoc deterministic stress was added after the original confirmation; it establishes a harmful law inside the certified class, not its frequency in practice.

The direct violation rate is 44.4% (exact 95% interval [27.9, 61.9]). All 16 failures are Waterbirds releases; the 20 BiasBios releases remain safe. The median maximum structural total-variation radius is .279, while the median ℓ_1 distance beyond the direct confidence region is .368 after deterministic resolution of an exact attacker tie. Across the registered error thresholds .30, .35, .40, .45, .49, direct violations are 18/20, 4/20, 16/36, 20/40, and 23/43; the candidate-matched strict MOSAIC rule has zero violations at every

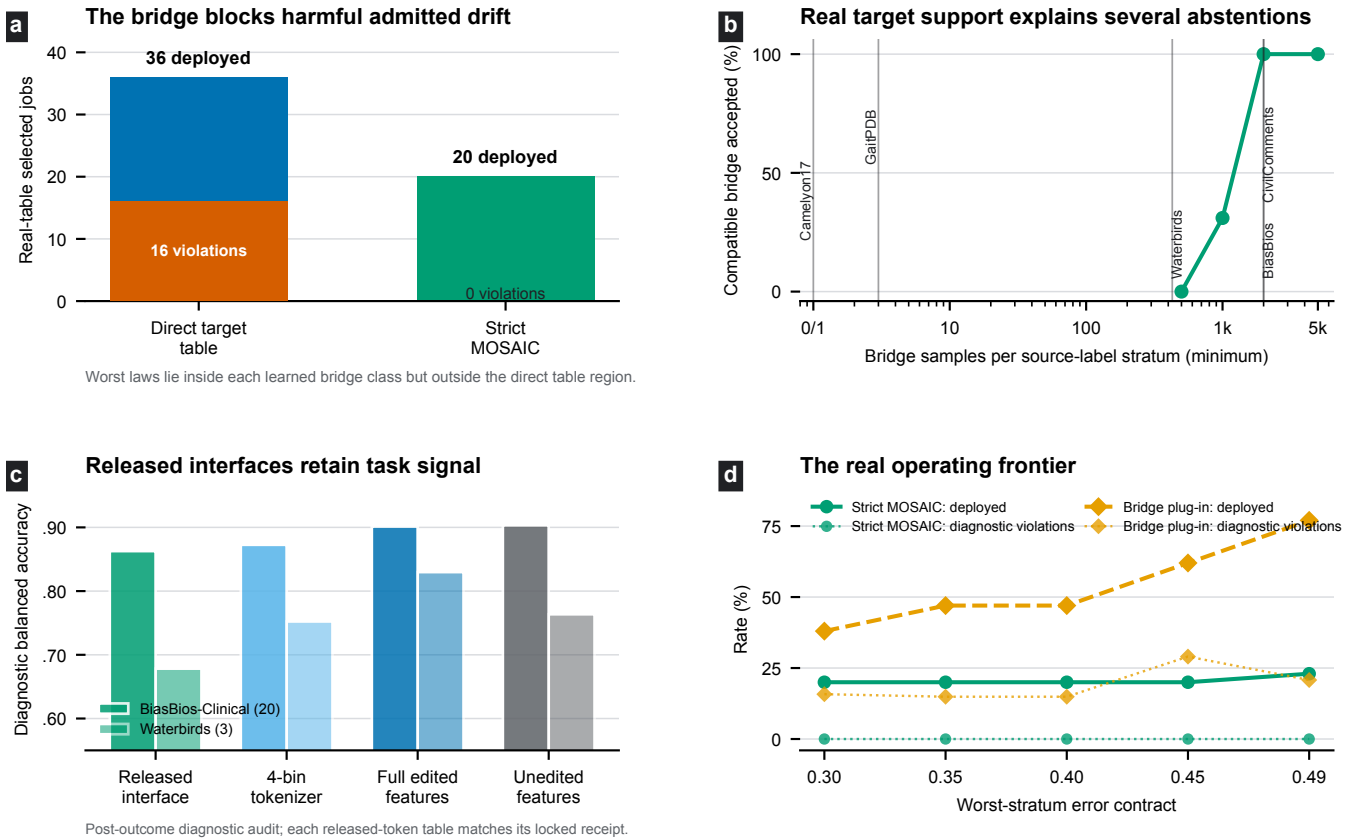


Figure 1: Additional audited evidence. (a) In a post-hoc stress test under real-table-anchored laws in the learned bridge class, the direct rule violates 16/36 accepted contracts; MOSAIC releases 20 safely and abstains on those 16 cases. (b) Real bridge stratum support is overlaid on the locked bridge-power curve. (c) The released interfaces retain task signal on untouched diagnostics. (d) Deployment and observed diagnostic-violation rates across the registered utility frontier.

threshold.

Complementary direct-region scope stress. The preceding construction is intentionally favorable to the bridge: it moves to a law inside the learned bridge class and outside the direct target region. We therefore also audit the opposite geometry using the two invalid target laws from the prelocked 12,000-table bridge-misspecification study. Their population retained masses are .775 and .50, below the declared .80 minimum. Among the 8,000 invalid rows, the known target law remains inside every rowwise direct target region in 7,983 cases (3,987 underdeclared-contamination and 3,996 source-specific-transform cases). MOSAIC’s membership gate abstains on all 8,000 rows, including all 7,983 direct-region cases.

This is a scope check, not a claim that the direct target-table rule fails inside its own confidence region: its sampling statement remains intact there. The comparison instead makes the two declared sets visible and shows the bridge’s operational tripwire when its structural assumption is violated. The frozen receipt records every rowwise membership calculation.

Target-law scope	Direct-region rows	MOSAIC abstains	ab-
Bridge-admitted, outside direct region	0/36	16/36	
Direct-region, bridge-invalid	7,983/8,000	8,000/8,000	

Table 9: Complementary scope cells. The first is the post-hoc real-table stress; the second is a deterministic recount of prelocked bridge-invalid tables.

Released-interface utility. Table 10 gives the values behind Fig. 1(c). Intervals are descriptive two-sided 95% t intervals over released jobs. Every diagnostic token table matches its locked receipt.

19 Finite-Alphabet Scaling

The post-review scaling study crosses five fine alphabets and three source counts with five sampled tables per cell, for 75 optimizations at $n = 2000$ per source-label stratum. All cells use two labels, two released tokens, source advantage .35, utility error .40, and residual contamination .05. Figure 2 reports every cell. All 75 releases certify at the registered .40 contract. Tightening the error contract reveals the expected

Interface evaluated	BiasBios-Clinical ($n = 20$)	Waterbirds ($n = 3$)
Released stochastic interface	.863 [.857, .868]	.678 [.636, .721]
Four-bin tokenizer	.873 [.867, .878]	.752 [.673, .831]
Full edited features	.901 [.898, .904]	.830 [.811, .849]
Unedited features	.903 [.902, .905]	.764 [.745, .783]

Table 10: Mean diagnostic balanced accuracy and 95% interval. The released interface is compact and task-specific; it is not a reusable embedding.

retention frontier as the multinomial radius grows with K .

20 Natural Multi-State ACS Confirmation

The data lock authenticates 12 processed stores before execution. The 60 receipts contain 1,560 official-candidate rows across two fine alphabets; 0 candidate rows record optimization failures. Bridge and diagnostic sets use disjoint target-state PUMAs in every receipt. Tables below report all tasks and states at the registered .40 contract. Primary selected-method counts are $K = 4$ MOSAIC: INLP=2, LEACE=8, MANCE++=4, R-LACE=1; $K = 4$ direct: INLP=3, Identity=13, LEACE=16, MANCE++=1, R-LACE=4, TaCo=10; $K = 8$ MOSAIC: none; $K = 8$ direct: INLP=3, Identity=11, LEACE=15, MANCE++=2, R-LACE=4, TaCo=12. Independent exact replay checks 60 receipts, 1,560 bridge certificates, 3,120 outward release bounds, and 1,200 stored selections with 0 failures.

21 Universal-Region Optimality and Deployment Extensions

21.1 A matching certification lower bound

Consider two equally likely sources. Source zero emits $\text{Ber}(1/2)$ and source one emits $\text{Ber}(1/2 + \theta)$. Their normalized Bayes source advantage is exactly θ .

Theorem 18 (Binary embedded lower bound). *Fix a contract τ , margin γ , soundness error δ , and power error β , where $0 < \gamma < \min\{\tau, 1/2 - \tau\}$ and $\delta + \beta < 1/2$. Any procedure that certifies $\theta = \tau - \gamma$ with probability at least $1 - \beta$ and certifies $\theta = \tau + \gamma$ with probability at most δ requires*

$$n \geq \frac{\log(1/[2(\delta + \beta)])}{\text{KL}(\text{Ber}(1/2 + \tau - \gamma) \| \text{Ber}(1/2 + \tau + \gamma))} \quad (36)$$

observations from the informative source stratum.

Proof. The certification decision is a test between the safe and unsafe experiments. Its two errors sum to at most $\delta + \beta$. The Bretagnolle–Huber testing inequality lower-bounds that sum by $\frac{1}{2} \exp[-n \text{KL}(P_{\text{safe}} \| P_{\text{unsafe}})]$. The other source has the same law under both experiments and contributes no KL. Rearranging proves Eq. (36). $\square$

For τ bounded away from $1/2$, this is $\Omega(\log(1/(\delta + \beta))/\gamma^2)$. More generally, consider protocols required to return a distribution-uniform ℓ_1 region that remains valid for every bounded functional selected after viewing the same K -category table. The dual identity $\|p - q\|_1 = \max_{v \in \{-1, 1\}^K} v^\top (p - q)$ reduces simultaneous support control to unrestricted multinomial ℓ_1 estimation. Standard

multinomial packing and the binary subexperiment above give

$$n = \Omega\left(\frac{K + \log(1/\delta)}{\gamma^2}\right). \quad (37)$$

The Weissman radius used by MOSAIC gives the matching upper rate. This optimality statement concerns universal same-table regions; a test of one fixed scalar functional can be easier.

21.2 Exact residual support and the infinite-sample floor

Fix one label, common transform T , reference laws p_s , release channel M , and residual budget η , so

$$q_s = (1 - \eta)p_s T + \eta r_s,$$

where each r_s is unrestricted. For attacker assignment a , let $v_{a,s}$ be its score on the fine alphabet and $R_a(q) = G^{-1} \sum_s q_s^\top v_{a,s}$.

Theorem 19 (Sharp residual floor). *For every fixed attacker assignment and task-loss vector u ,*

$$\sup_{r_{1:G}} R_a(q) = \frac{1 - \eta}{G} \sum_s p_s^\top T v_{a,s} + \frac{\eta}{G} \sum_s \max_c v_{a,s}(c), \quad (38)$$

$$\sup_r q_s^\top u = (1 - \eta)p_s^\top T u + \eta \max_c u(c). \quad (39)$$

These terms persist at zero reference radius and no uniformly valid certificate over the stated bridge class can make them smaller. If two common released laws coincide and M has binary Dobrushin coefficient α , the class contains source advantage exactly $\eta\alpha$.

Proof. Each objective is linear in every residual law, hence maximized at a simplex vertex placing all mass on a highest-scoring fine token. Those vertices belong to the registered bridge class and attain both displays. For the final claim, choose source residuals on two channel rows attaining total variation α ; their common component cancels and the residual contribution has total variation $\eta\alpha$. $\square$

This separates statistical uncertainty from declared shift breadth. Tighter multinomial regions can reduce the first, but only a narrower bridge class or a more contractive channel can reduce the second. A locked census recomputes all 35 primary BiasBios and ACS releases in four modes: full, zero reference radius, zero residual, and both zero. Residual shift exceeds sampling in all 35 jobs for both source and utility. Median source increments are .1914 versus 4.01×10^{-8} ;

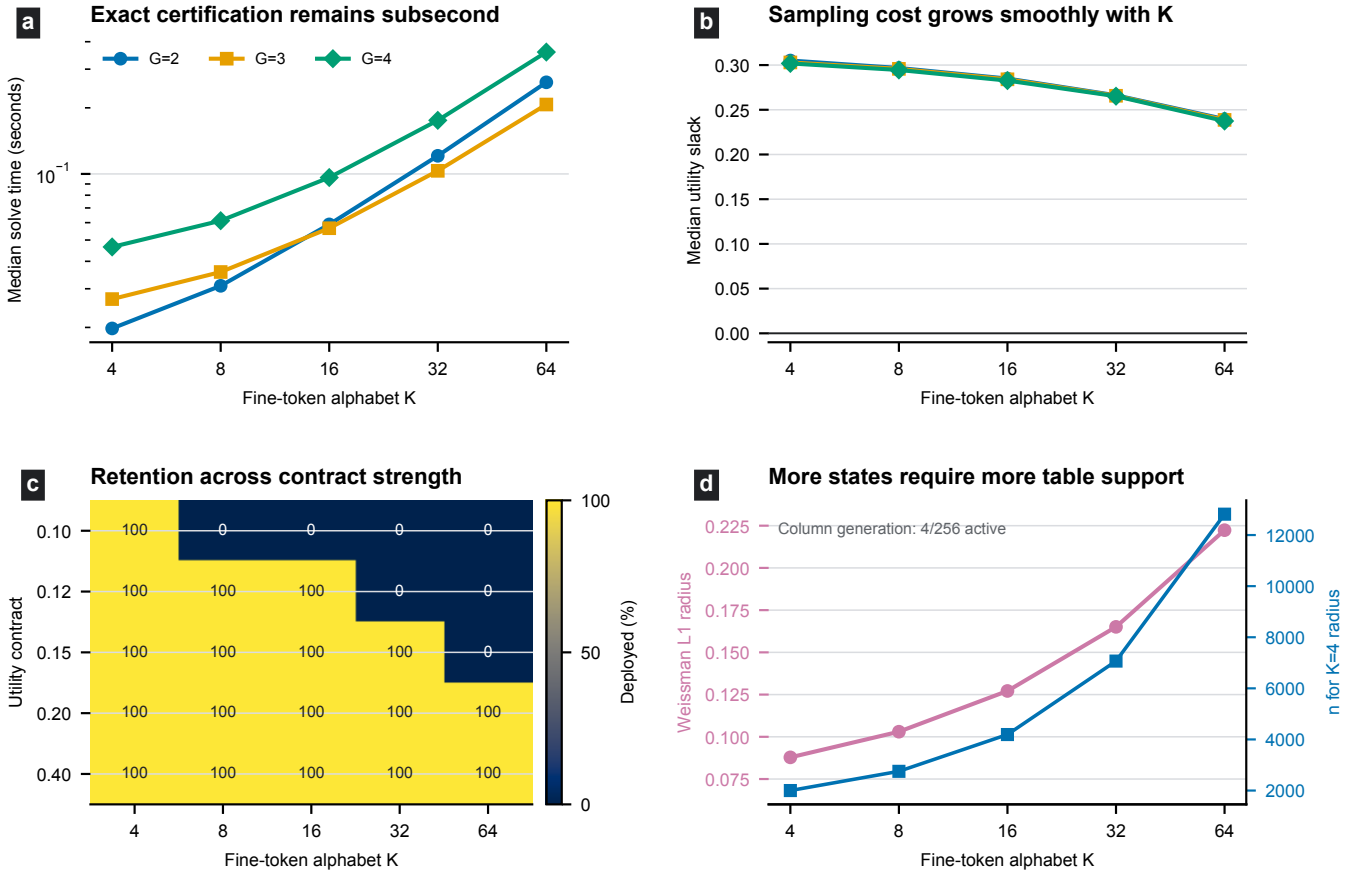


Figure 2: Finite-alphabet scaling over five independent tables per cell. (a) Median wall-clock optimization. (b) Median utility slack at the .40 contract. (c) Retention averaged over $G = 2, 3, 4$ at tighter contracts. (d) The Weissman radius and per-stratum sample size needed to recover the $K = 4$ radius for $G = 4$. A $G = L = 4$ constraint-generation stress uses 4 of 256 possible attacker assignments.

median utility increments are .1377 versus 8.26×10^{-9} . An independent replay cross-checks every frozen release and all summary statistics.

21.3 Bounded correlated sessions

Let a source be associated with r fine items $C_{1:r}$ having an arbitrary joint source-conditional law. Registered channels M_i release Z_i , with independent mechanism randomness conditional on the complete fine tuple. Define

$$M_{[r]}(c_{1:r}, z_{1:r}) = \prod_{i=1}^r M_i(c_i, z_i).$$

Theorem 20 (Exact bounded-session reduction). *Applying the MOSAIC envelope to a simultaneous confidence region for the joint fine-tuple law and channel $M_{[r]}$ gives an exact universal-attacker certificate for the complete transcript. No independence assumption is made on $C_{1:r}$.*

Proof. Conditional on the tuple, the displayed product is exactly the transcript law. Marginalizing the unknown joint fine law through this row-stochastic matrix produces the attacker's experiment. The adaptive-envelope proof requires

only a finite source law and a row-stochastic channel, so it applies unchanged. $\square$

Write $\alpha_i = \max_{c, c'} \text{TV}(M_i(c, \cdot), M_i(c', \cdot))$. Maximal coupling of each item channel gives the scalable bound

$$\alpha(M_{[r]}) \leq 1 - \prod_{i=1}^r (1 - \alpha_i). \quad (40)$$

Indeed, the transcript differs only if at least one item coupling differs. The coupling characterization of total variation proves Eq. (40); the normalized Bayes source advantage for any number of sources is no larger than the same quantity. Arbitrary registered adaptive session mechanisms are also covered after compilation into one conditional transcript channel.

21.4 Anytime-valid recertification

For categorical counts N_n and a fixed Dirichlet mixing law π , define

$$E_n(p) = \frac{\int \prod_j q_j^{N_{n,j}} \pi(dq)}{\prod_j p_j^{N_{n,j}}}. \quad (41)$$

K	Median time (s)	ℓ_1 radius	Utility slack	Retained at .40	n for $K = 4$ radius
4	.047	.0878	.3019	5/5	2,000
8	.062	.1030	.2945	5/5	2,752
16	.097	.1271	.2826	5/5	4,192
32	.174	.1651	.2652	5/5	7,067
64	.360	.2223	.2374	5/5	12,817

Table 11: $G = 4$ scaling summary; the complete artifact also contains $G = 2, 3$. The synthetic tables isolate computation and confidence-radius scaling.

K	Rule	Error cap	Releases	Diagnostic viol.
4	MOSAIC	0.30	0/60	–
4	MOSAIC	0.35	12/60	0/12
4	MOSAIC	0.40	15/60	0/15
4	MOSAIC	0.45	46/60	0/46
4	MOSAIC	0.49	57/60	0/57
4	Direct table	0.30	20/60	0/20
4	Direct table	0.35	40/60	1/40
4	Direct table	0.40	47/60	0/47
4	Direct table	0.45	60/60	0/60
4	Direct table	0.49	60/60	0/60
8	MOSAIC	0.30	0/60	–
8	MOSAIC	0.35	0/60	–
8	MOSAIC	0.40	0/60	–
8	MOSAIC	0.45	0/60	–
8	MOSAIC	0.49	1/60	0/1
8	Direct table	0.30	22/60	0/22
8	Direct table	0.35	40/60	0/40
8	Direct table	0.40	47/60	0/47
8	Direct table	0.45	60/60	0/60
8	Direct table	0.49	60/60	0/60

Table 12: Complete registered utility-threshold frontier. No cell is omitted.

Task	K	MOSAIC rel.	MOSAIC viol.	Direct rel.	Direct viol.
employment	4	13/20	0/13	20/20	0/20
employment	8	0/20	–	20/20	0/20
income	4	2/20	0/2	20/20	0/20
income	8	0/20	–	20/20	0/20
public coverage	4	0/20	–	7/20	0/7
public coverage	8	0/20	–	7/20	0/7

Table 13: Primary-contract results by task. Violations are measured only on released jobs with complete held-out PUMA support.

Theorem 21 (Dirichlet-mixture confidence sequence). *Under law p , $E_n(p)$ is a nonnegative martingale with initial value one. Consequently $\mathcal{C}_n(\delta) = \{p : E_n(p) < 1/\delta\}$ contains the true law simultaneously for every $n \geq 0$ with probability at least $1 - \delta$, including at arbitrary stopping times.*

Proof. For every fixed q , its likelihood ratio against p is a nonnegative martingale. Integrating against a fixed mixing law preserves that property. Ville’s inequality bounds $\Pr_p(\sup_n E_n(p) \geq 1/\delta)$ by δ . $\square$

For Dirichlet parameter a and $\hat{p} = N/n$,

$$\log E_n(p) = \log E_n(\hat{p}) + n \text{KL}(\hat{p} \| p).$$

Target	K	MOSAIC rel.	MOSAIC viol.	Direct rel.	Direct viol.
FL	4	4/15	0/4	14/15	0/14
FL	8	0/15	–	14/15	0/14
IL	4	2/15	0/2	13/15	0/13
IL	8	0/15	–	13/15	0/13
NY	4	5/15	0/5	10/15	0/10
NY	8	0/15	–	10/15	0/10
WA	4	4/15	0/4	10/15	0/10
WA	8	0/15	–	10/15	0/10

Table 14: Primary-contract results by target. Violations are measured only on released jobs with complete held-out PUMA support.

This identity gives an exact KL confidence sequence; Pinsker’s inequality provides the implemented ℓ_1 outer radius. Allocating δ across registered strata yields one event covering every MOSAIC channel selected at every recertification time.

21.5 Continuous release classes

Let $q : x \mapsto \Delta_L$ range over a registered class with a finite ε -net of size N in $\sup_x \|q(x) - q_j(x)\|_1$. Coordinate Hoeffding bounds and two approximation terms give, simultaneously for every same-table selected q ,

$$\left\| \mathbb{E}q(X) - \frac{1}{n} \sum_{i=1}^n q(X_i) \right\|_1 \leq L \sqrt{\frac{\log(2LN/\delta)}{2n}} + 2\varepsilon. \quad (42)$$

Passing these output-law regions through the identity channel gives the finite-output MOSAIC certificate without first exposing a hard fine token. For all thresholds $q_t(x) = (\mathbf{1}\{x \leq t\}, \mathbf{1}\{x > t\})$, DKW removes the finite net:

$$\sup_t \left\| \mathbb{E}q_t(X) - n^{-1} \sum_i q_t(X_i) \right\|_1 \leq 2\sqrt{\frac{\log(2/\delta)}{2n}}.$$

Thus a threshold may be selected from the same audit sample with no grid or threshold-count correction.

21.6 Token-dependent real-proxy calibration

Let R be an imputed source and allow $Q_y(s, c, r) = \Pr(R = r \mid S = s, C = c, Y = y)$ to vary with the fine token. If $x_y(s, c)$ is the latent joint law, the observed proxy-token law is the linear image $r_y(r, c) = \sum_s Q_y(s, c, r)x_y(s, c)$.

Theorem 22 (Token-dependent proxy bridge). *Suppose simultaneous regions cover the observed r_y and every calibrated row $Q_y(s, c, \cdot)$. Replacing the proxy observation*

matrix by this token-dependent tensor and enlarging the observed-law radius by the largest calibrated row radius contains the true latent law. The sharp conditional token bounds follow from the same Charnes–Cooper LPs; an empty calibration cell yields maximal uncertainty and abstention.

Proof. Writing A_Q for the induced linear map, $\|(A_Q - A_{\hat{Q}})x\|_1$ is bounded by the largest calibrated row error, since it is a convex combination of row errors. The triangle inequality puts $A_{\hat{Q}}x$ in the enlarged observed region. Conditional extrema are linear-fractional functions over this polytope, so the existing Charnes–Cooper reduction is exact. $\square$

21.7 A sharp privacy–utility conflict

For a balanced binary source, write $\kappa = \Pr(Y \neq S)$ after fixing the registered identification of task and source labels.

Theorem 23 (Source–task conflict). *Every released observation and decoder with task error e_Y and source advantage at most τ satisfies*

$$e_Y \geq \max\{0, (1 - \tau)/2 - \kappa\}.$$

The same bound applies to worst-label error, and it is sharp when $Y = S$.

Proof. Use the task decoder as a source decoder. Its source error is at most $e_Y + \kappa$, so the Bayes source error is no larger. Source advantage is therefore at least $1 - 2(e_Y + \kappa)$. Rearrangement proves the bound. For $Y = S$, a binary symmetric channel with crossover $(1 - \tau)/2$ attains equality. $\square$

22 Path-to-Deployment Confirmations

22.1 Locked theory implementation study

The extension implementation was locked before execution; the artifact manifest records SHA-256 prefix `99ccc7534bd1d194` and the complete digest. Across 1,000 optional-stopping runs for each $K \in \{2, 4, 8\}$, anytime false exclusion occurs in 5, 2, and 0 runs, respectively, below the registered familywise .05 level. Five product-channel rows match exact enumeration, and 500 continuous-threshold selections satisfy Eq. (42). The audit independently reconstructs every check and reports pass.

22.2 Natural image-origin shift

The frozen CINIC-10 store contains 48,000 file-provenanced images from four classes, equally allocated across CIFAR-10 and ImageNet origin and across construction, reference, bridge, and diagnostic roles. ResNet-18 penultimate features are fixed before the audit. Five pre-outcome seeds compare Identity and official closed-form LEACE and release 5/5 interfaces without a diagnostic violation. After that result, a disclosed sequential power extension locks 35 new seeds and reruns all 40 under a stricter 40-seed familywise allocation. MOSAIC releases 40/40 with 0/40 violations. Selected certified worst error ranges from .141 to .213; diagnostic source advantage ranges from .016 to .038. The released channel costs a median .60 balanced-accuracy points relative to its selected four-token task interface.

22.3 A real estimated proxy

A locked ACSIncome follow-up trains a source imputer from geography, birthplace, ancestry, Hispanic origin, language, nativity, and citizenship on a disjoint partition; balanced accuracy is .8413. True-source calibration data estimate a label- and token-dependent confusion tensor, while the target certificate sees proxy labels. The follow-up adds simultaneous source-mass intervals to the latent-law polytope and uses its exact ℓ_1 Chebyshev center. This constraint was developed on a disclosed earlier split that abstained, then locked under seed 20270725 before the new diagnostic outcome.

At 19,566 and 39,133 calibration rows, maximum conditional radii .5534 and .4771 force abstention. At 58,700 and 78,267 rows, radii .4126 and .3669 permit release. The primary full-data certificate bounds source advantage by .2175/.2790 and worst error by .3691. The untouched diagnostic values are .0590 and .2769, with released expected balanced accuracy .7729 versus .8144 for the unedited task score. The auditor independently reruns the optimizer, verifies all 28 checks, and accounts for the complete .05 familywise budget. ACS lacks surnames, so the experiment is not labeled BISG.

22.4 Matched local differential privacy

For every primary MOSAIC release in BiasBios and ACS, a matched baseline uses the same four-token input, binary output, bridge class, radii, and .40 utility contract. Four-ary randomized response keeps the registered task bit with probability .675, giving $\epsilon_{\text{LDP}} = .7309$ and Dobrushin coefficient .35, exactly the source-advantage threshold. We enumerate all 16 deterministic fine-token task maps before applying that fixed channel.

Randomized response meets the stronger input-local-DP guarantee, but releases 0/35 interfaces because its certified task error exceeds .40. MOSAIC releases 35/35 under its narrower source-inference contract and has lower certified error in every job. The median, minimum, and maximum error penalties for randomized response are .1554, .0939, and .1710. The independent audit verifies all 35 local-DP channels, all source receipts, and the complete summary.

22.5 A prospective natural direct-rule failure

The original ACS geographic study used 2018 California reference data and held-out 2018 target states. A later pandemic-panel analysis identified two fixed public-coverage interfaces that direct certification released while MOSAIC abstained. The pattern was discovered in 2021 and persisted in 2022. Before accessing 2023, a lock fixed both channels, their task decoders, Florida as target, the .40 utility threshold, the exact Census archive member and checksum, and a familywise Hoeffding lower bound over eight strata.

Both interfaces violate the threshold empirically in 27,772 Florida rows. For the TaCo interface at seed 1401, worst error is .42453 and the familywise lower bound is .40542, confirming a contract violation. The R-LACE interface has error .40812 and lower bound .38900, so it is an empirical but not familywise-confirmed violation. MOSAIC had

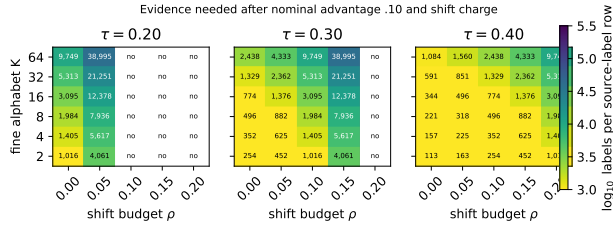


Figure 3: Sufficient labels per source-label stratum as a function of alphabet size, source contract, and charged shift budget. Blank cells are population-infeasible before sampling. The machine-readable chart contains all 120 registered configurations.

abstained on both 2018 decisions. A separate 52-check auditor reconstructs the 2018 reference functionals, verifies the streamed 2023 archive hash, and recomputes every bound. This is a prospective confirmation of a previously identified failure mode, not a first-look discovery.

22.6 Designer chart

For example, with four strata, nominal source advantage .10, contract $\tau = .20$, and no shift charge, the sufficient counts are 1,016, 1,405, and 1,984 for $K = 2, 4, 8$. A .05 shift charge reduces the statistical margin by half and raises them to 4,061, 5,617, and 7,936. At shift charge .10, the population bound already meets the contract and no finite sample can create strict slack.

22.7 Commensurate official FARE comparison

Official FARE commit 89cblb66ed268c16 trains eight `fair_gini_eo` tree representations on the same raw ACSIncome inputs and five-way partition as the real-proxy study. Maximum leaves are $\{2, 4\}$ and $\alpha \in \{.80, .90, .95, .99\}$; every terminal leaf is judged by the same token-dependent proxy calibration and .35 source, .40 task contracts. All eight trees realize two leaves and abstain. Their worst source bound is 1.0 in one label and their worst conditional task bound is 1.0; no candidate is released and there is no false acceptance. This is an end-to-end representation comparison under one commensurate release contract. It does not reinterpret FARE’s original demographic-parity theorem as a source-inference guarantee.

23 Mechanized Core and Auditor-Facing Tool

The Lean 4 project under `research/formal/MosaicFormal` machine-checks the adaptive-selection implication, familywise lifting, and the discrete bridge residual construction used by Theorems 1–2. Its build receipt records Lean 4.32.1 and theorem names `adaptive_exact_envelope_sound`, `adaptive_family_envelope_sound`, `bridge_residual_reconstruct`, and `adaptive_bridge_residual_reconstruct`. Concentration inequalities, LP duality, and floating-point

solver correctness remain separately audited by the 585-check implementation-independent replay.

The installable `mosaic-certified-release` package exposes `fit`, `certify`, and `release_or_abstain`. The `mosaic-audit` command validates a serialized certificate and emits a versioned JSON or text report. Runtime monitoring records evidence age, contract version, support state, and recertification decisions; failed or expired evidence closes the release. A stand-alone replication packet specifies exact commands, expected hashes, and discrepancy reporting for an evaluator outside the project. No external replication is claimed in this submission.

24 Registered Qwen Pilot Boundary

A prewritten unlocked pilot evaluated six Qwen2.5-1.5B-Instruct hidden-state interfaces on 12,264 CivilComments-WILDS examples. Integer dataset IDs divisible by four formed the pilot population; every later confirmation ID was excluded. The familywise .05 budget covered three fixed pooling rules, $K \in \{4, 8\}$, and both reference and bridge tables. The go rule required complete strata, minimum retained mass at least .50, a nonconstant release, source advantage at most .35, and certified worst error at most .49.

No candidate passed. The layer-14 and final-layer mean-pooled $K = 4$ channels were nonconstant, with retained masses .6154 and .6557, but their certified worst errors were .492292 and .490117. Every $K = 8$ candidate and the final-token $K = 4$ candidate returned a constant release with error .5. An independent audit verifies all six candidates, all role-label-source counts, zero confirmation-ID overlap, every manifest hash, and the exact stopping rule. Consequently no temporal confirmation was registered and the main paper makes no LLM-interface claim.

25 Reproducibility Manifest

MOSAIC optimization, confirmations, and audits ran on an Apple M4 system with 10 CPU cores and 16 GB unified memory under macOS 26.2 (arm64). The locked Python environment was Python 3.12.13 with NumPy 2.5.1, SciPy 1.18.0, scikit-learn 1.9.0, PyTorch 2.13.0, and the HiGHS LP/MILP backend exposed by SciPy. No GPU is required once the registered feature stores have been constructed; synthetic confirmation and every replay are CPU-only.

The named package reports the exact preregistration hash, Git commit, software versions, and artifact hashes. The anonymous package retains the same internal identifiers while replacing public authorship and repository metadata as required by double-blind review. Raw synthetic receipts contain no private or human-subject data. Real-domain experiments, if included, retain dataset terms, official split definitions, preprocessing hashes, and method-specific code provenance in a separate manifest.

The repository is also installable as `mosaic-certified-release`. Its public path is `fit`, `certify`, then `release_or_abstain`; the runtime persists one sampled token per item after certification and fails closed after missing support or invalid

Artifact group	Checksum index
Main and supplement PDFs	V12 release SHA-256 manifest
Locked protocols and reports	Anonymous code/data manifest
Real compact receipts and audits	Anonymous code/data manifest
Post-review stress, scaling, natural shift, and Qwen pilot	V12 release SHA-256 manifest

Table 15: Single-entry checksum map. Full digests are stored once in the two machine-readable release manifests rather than repeated in prose.

recertification. The anonymous archive contains the same package with sanitized authorship metadata and focused end-to-end tests.